



Castor Oil Cake: A Sustainable Organic Nutrient Source for Sericultural Host Plants: A Review

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ABSTRACT

The rapidly increasing environmental degradation caused by indiscriminate use of synthetic fertilizers has pivoted our focus towards sustainable and organic nutrient management agricultural strategies. Castor oil cake (COC), a nutrient-rich by-product generated during castor oil extraction, has emerged as a promising organic nutrient source due to its high nitrogen (4-6%), phosphorus (1-2%) and potassium content (1-2%). Although COC contains toxic compounds such as ricin, which limits its use as livestock feed, it has been widely investigated as a soil amendment for improving soil fertility and crop productivity. This review aims to evaluate the potential of castor oil cake as an organic nutrient source for sericultural host plants, particularly castor (*Ricinus communis* L.), the primary host plant of the eri silkworm (*Samia ricini* Donovan). Relevant literature published on COC and related organic nutrient management practices was critically reviewed to assess its effect on soil properties, microbial activity, plant growth and its potential relevance to sustainable ericulture. The reviewed studies indicated that optimum COC application enhances soil microbial activity, improves moisture retention and promotes nutrient availability. Apart from biomass accumulation, COC has also been reported to improve the physiological resilience to salt stress and enhance phytochemical production in crops. In the context of ericulture, COC-driven soil enrichment produces superior foliage quality, thereby supporting improved silkworm growth and cocoon production. By repurposing industrial waste into an efficient nutrient source, COC offers a sustainable pathway to mitigate chemical reliance and boost soil health. However, direct evidence relating COC application to eri silkworm performance remains limited. The utilization of castor oil cake as an organic fertilizer offers a practical strategy for recycling an agro-industrial by-product into a valuable nutrient source. Its application has the potentiality to reduce reliance on chemical fertilizers, improve soil health, enhance castor growth and productivity and promote environmentally sustainable nutrient management practices in sericulture. COC has the potentiality to enhance soil productivity and thereby improving eri silkworm productivity through improved host plant nutrition. Although its impact has been well documented in a number of crops, further research is required to establish standard application practices and its effects on eri silkworm growth, cocoon yield and economic returns. Therefore, this review aims to highlight the potentiality of castor oil cake as an organic fertilizer for eri silkworm host plants to improve overall soil health, host plant productivity and environmentally sustainable sericultural practices.

Key words: Castor oil cake, Castor plants, Eri silkworm, Organic agriculture, Sericulture, Soil health, Sustainability.

The environmental degradation caused by indiscriminate use of high-analysis synthetic fertilizers has become a critical global concern. In sericulture, where leaf quality directly determines silkworm growth and cocoon production, maintaining soil fertility through environmentally sustainable nutrient management is particularly important (Nandan and Ashwini, 2026). This has led to the prioritization of other sustainable and eco-friendly agricultural approaches. In this scenario, organic fertilizers can be a vital source of nutrients for proper crop growth, development, productivity as well as for maintaining good soil health (Jana *et al.*, 2024; Begum *et al.*, 2025). Among these, one such organic input that could serve as a promising organic fertilizer is castor oil cake, a by-product generated from castor seeds during its oil extraction process. It possesses a rich nutrient profile and provides additional agronomic benefits (Lima *et al.*, 2011). India is one of the major leading producers of castor seeds worldwide, cultivating across 0.89 million hectares. It records a yield of 3.76 MT annually with productivity of 1928

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kg ha⁻¹ (Directorate of Oilseeds Development, 2025). The production of castor in India has been observed to follow a trend of progressive growth since 2016-17, indicating stable and sustained improvement in recent years (Fig 1).

The country's castor oil export ranges to 80 % securing a strong position in the international market. The castor

seeds contain almost 48% oil, out of which 42% of oil can be extracted and some amount remained with the castor cake. Each tonne of processed castor seeds produced approximately 550 kg of castor oil cake (Annongu and Joseph, 2010), while nearly 2.4 tonnes of oil cake are obtained from the extraction of one tonne of castor oil (Lima *et al.*, 2011) through expeller press and solvent extraction methods. To effectively manage such a huge quantity of oil cake produced every year, the potential use of oil cake is the need of the hour. This large-scale production of castor seeds in India generates a substantial quantity of castor oil cake, creating an opportunity for its utilization as an organic nutrient source. This is particularly relevant to eri sericulture because castor is the primary host plant of eri silkworm (*Samia ricini*) and the productivity of castor directly influences leaf availability and quality for silkworm rearing (Banerjee *et al.*, 2026). The effective utilization of such nutrient-rich organic residues as a strategic supplement to synthetic fertilizers can offer a sustainable pathway for nutrient management in agriculture. In addition to supplying essential nutrients, they also serve as catalysts for improving soil properties and stimulating the microbial activity (Shaji *et al.*, 2021). This is evident as the nutritional structure of castor oil cake consists of considerable amount of nitrogen (6.4%), phosphorus (2.55%) and potassium (1%) (Deshmukh *et al.*, 2022). Not only this, it is also found to have presence of other trace elements (Gadhesariya and Malam, 2021). Although castor oil cake possesses excellent fertilizer value, its utilization as livestock feed is prohibited due to the presence of toxic compounds such as ricin and ricinine (Mondal *et al.*, 2019). This limitation serves as a significant opportunity to repurpose this by-product within eco-friendly farming practices while reducing the dependency on synthetic fertilizers. Therefore, its potentiality as organic fertilizer could be explored due to presence of both macro as well as micro-nutrients.

The rearing of Eri silkworm (*Samia ricini*) largely depends on the availability and quality of its primary host plants, mainly castor (*Ricinus communis* L.). The nutritional

quality of primary host plants directly influences the larval growth, cocoon yield and silk quality. While the beneficial effects of COC in enhancing soil fertility and crop productivity are well documented, the information regarding its specific role in castor cultivation for eri sericulture is limited and scattered. Particularly, evidence linking COC application with castor leaf quality, nutritional composition and subsequent performance of eri silkworm is scarce. Consequently, identifying sustainable nutrient sources that can enhance the soil health and quality of the host plants for environmentally sustainable sericulture is becoming more important. Conventionally, chemical fertilizers have been widely used to improve the crop growth and leaf productivity; however, continuous application of high analysed fertilizers often leads to soil degradation, nutrient imbalance and reduced soil-microbial activity. In recent years, these conventional approaches are being replaced by organic alternatives and castor oil cake (COC) emerges as a promising organic fertilizer. As mentioned earlier, the COC offers a multifaceted approach to soil health management. It enhances the soil physical properties, additionally stimulating microbial populations and increasing moisture retention. This helps ensure optimum growth of castor plants with the gradual and sustained release of nutrients, facilitating constant nourishment. Eventually, improved soil fertility and foliage quality through the application of COC has the potential to positively influence silkworm growth and cocoon production; however, direct evidence in eri sericulture remains limited. Therefore, this review critically discusses the nutrient composition of COC, its impact on soil physical, chemical and biological properties, its effect on plant growth and productivity and its potential relevance to sericultural host plants, while highlighting the major challenges and future research priorities for its effective utilization in sustainable eri sericulture.

Origin

Castor (*Ricinus communis* L.) belongs to the family Euphorbiaceae. It comes under the genus *Ricinus* and

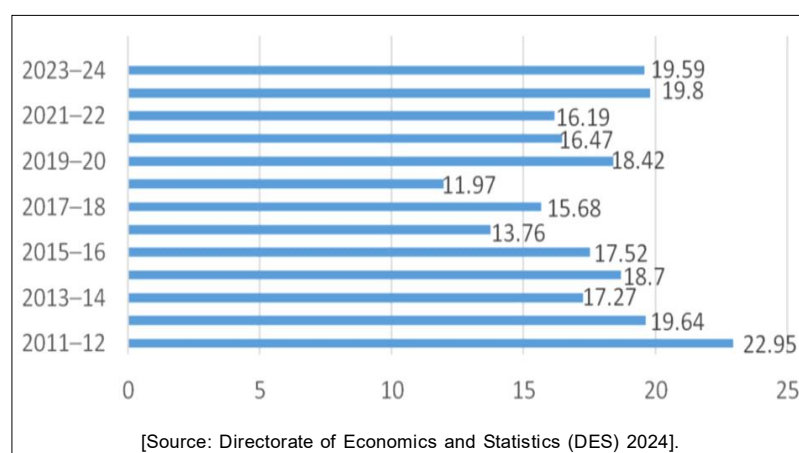


Fig 1: Production (lakh tonnes) of castor in India.

sub-tribe Riciniinae. Its origin has been traced back to East Africa, particularly Ethiopia, where its domestication is believed to have begun approximately 3,200 years ago (Xu *et al.*, 2021). The species exhibits remarkable ecological resilience following its African origins and helps proliferate across diverse global climates (Landoni *et al.*, 2023). Fig 2 represents a diagrammatic representation of its origin. In India, castor is cultivated extensively due to its wide adaptability across all geographic regions. Maximum area in the country is occupied by Gujarat (0.78 lakh hectares), followed by Rajasthan with 0.35 lakh hectares. Apart from these Andhra Pradesh, Karnataka and Odisha also contribute to a lesser extent (Anonymous, 2024). In addition to the soil enhancement properties, it holds immense industrial value due to its unique chemical composition. The seeds of castor contain approximately 90% ricinoleic acid. The presence of this unique fatty acid makes it an indispensable raw material for high-value utilization in industries including pharmaceuticals and cosmetics. On a global stage, India is one of the leading producers of castor seeds. In the current scenario, India contributes up to 85.02% (Kalamkar and Sharma, 2022). The country produces nearly 1.57 million tonnes of castor seeds from 0.83 million hectares (Kumar, 2020).

Production and availability of castor oil cake

Significance of castor oil cake as organic fertilizer

Organic fertilizers are known to be effective fertilizers to improve soil health and fertility, including its structure and physico-chemical properties. Among various organic nutrient sources, castor oil cake (COC) has emerged as a promising soil amendment due to its high nutrient content and favourable effects on soil properties. Although the presence of toxic compounds such as ricin and ricinine restricts its use as livestock feed, it can be efficiently utilized as an organic fertilizer. Candian *et al.*, (2024) stated that optimizing the application rate of COC between 3000 kg

ha⁻¹ and 3800 kg ha⁻¹ enhanced the plant growth and crop yields by accelerating the release of essential nutrients into the rhizosphere. In an experiment conducted on basil cultivation, it has been observed that the COC application mitigated deleterious effect of salt stress by facilitating osmoregulation, maintaining cell turgidity and improving photosynthetic efficiency (Sousa *et al.*, 2022). Furthermore, when integrated into diverse cropping systems, such as fennel cultivation, COC has been shown to improve essential oil content, whether applied as a sole amendment or in conjunction with inorganic fertilizers (Patel *et al.*, 2003).

To maximize the efficacy of castor oil cake and ensure the degradation of its inherent phytochemical toxins, both timing and dosage are critical parameters. Evidence suggested that COC should be incorporated into the soil at least three weeks before planting to facilitate the detoxification of harmful substances and allow for proper decomposition (Gupta *et al.*, 2004). Such pre-plant incorporation ensures a gradual release of nutrients during crop growth while maximizing any adverse effects associated with fresh cake application. The combined application of nitrogen (60 mg/kg soil) and castor cake (0.50%) has shown an increase in the phosphorus concentrations in crops like wheat from 0.36 to 0.42%. It demonstrates the potential of castor cake as a biological source of phosphorus. The advantages of COC have also been reported in other leguminous crops like peanuts. The application of COC has outperformed both conventional chemical fertilizers and traditional cow dung (Li *et al.*, 2024). The application of COC @ 10.08 t ha⁻¹, significantly enriched the soil by increasing organic carbon, nitrogen, phosphorus and potassium levels by 86.4%, 64.6%, 70.5% and 11.3%, respectively while modulating soil pH to a more favourable level. There has been a distinct bell curve relationship between dose and its response. Concentrations up to 4.5% of castor meal leads to increase in plant growth but an excessive dose leads to inhibitory effects. This positive response is attributed to the low C/N ratio of castor oil cake that helps in rapid mineralization process and accelerate nutrient uptake. All these characters contribute to making it an effective alternative for sustainable cultivation systems.

Despite its considerable fertilizer value, the use of COC requires precautions. The presence of toxic compounds like ricin and ricine makes COC unsuitable for livestock feed and requires careful handling during storage and application to avoid unnecessary exposure. Freshly applied COC may also cause phytotoxic effects; therefore, it should be incorporated into the soil at least three weeks prior to planting to allow sufficient decomposition and detoxification (Gupta *et al.*, 2004). The other major limitation associated with COC is the variation in its nutrient composition that depends on a number of factors including the technique used for oil extraction from the seeds, extent of decortication and other processing conditions. As a result, the fertilizer value as well as the appropriate amount of application of COC may differ depending on the batch used.

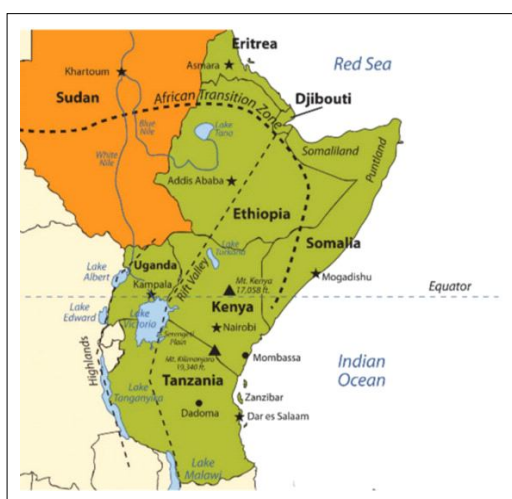


Fig 2: Castor is indigenous to eastern Africa, especially Ethiopia.

Nutrient dynamics of castor oilcake

Castor oil cake contains carbohydrates, proteins, fats, mineral matter and essential macro- and micronutrients. It is a significant organic by-product derived from the industrial extraction of oil from castor seeds (*Ricinus communis* L). It offers a balanced proportion of energy-rich and structural components. Studies suggest that the composition of cake consists of approximately 24.69% carbohydrates, 31.06% proteins and 19.40% fats with 11.10% mineral matters (Annongu and Joseph, 2010). However, the specific nutritional density can vary based on the processing method and whether the seeds are decorticated before oil extraction or not. The decorticated seed cake generally contains nutrient concentration on the higher end compared to the un-decorticated ones. In a study, it was recorded that decorticated cake has been found to contain approx. 24.88% carbohydrate, a significantly higher amino acid (35.43%) and an ether extract (crude fat) of 25.10% (Annongu and Joseph, 2010). It also recorded the presence of 7.14% total ash content. Castor oil cake also consists of higher essential macronutrients, including 4.5% to 6 % nitrogen, 1.50% phosphorus and 1.50% potassium. The physical composition of the cake also includes a moisture content of roughly 12% and a residual oil content of 3.5% to 8%. It helps in the slow and sustained release of nutrients during decomposition. Beyond these primary elements, castor oil cake serves as a vital reservoir for secondary and micronutrients that are often overlooked in traditional fertilization programs. Gadhesariya *et al.* (2021) highlight that the cake is a natural source of magnesium, calcium and sulphur, which are essential for chlorophyll synthesis and structural integrity. Furthermore, it supplies critical trace elements including iron, zinc, copper and manganese. The presence of these micronutrients helps catalyse various complex enzymatic and metabolic plant processes.

Effect of oil cakes on soil properties

Soil physical property

There is a profound influence of castor oil cake on soil physical properties due to its beneficial role as an organic soil amendment which not only improves soil fertility but also enhances several physical properties of the soil (Aytenew and Bore, 2020). The application of organic manures improves soil structure and texture resulting in better water retention capacity and increased porosity (Sajwan and Mishra, 2019). In a pot experiment conducted on cucumber and tomato, formulations containing neem (2.5% v/v) and niger cake (5% v/v) blended with coir pith and vermicompost achieved a bulk density of 0.28-0.29 g/cm³, total porosity of 83.37-84.63%, water-holding capacity of approximately 59%, electrical conductivity (EC) of 2.64-2.76 mS/cm and pH of 6.40-6.56. These results indicated a lightweight yet absorptive growing medium with adequate air space for root development of cucumber and tomato (Pratibha *et al.*, 2021). Similarly, the amendments

of castor cake at 4.5% (v/v) significantly stimulated castor plant growth in sandy media; however, growth diminished beyond this threshold, underscoring the importance of appropriate dosage. In clay soil, incorporation of 9% olive mill solid waste (by weight) enhanced penetration depth by 30% and accumulated water intake by 29%. While in sandy clay soil, the respective increases were 25% and 32% (Abu-Rumman, 2016). The different organic manures showed an impact on soil physical properties. However, the extent of amelioration depends upon the concentration of organic manures as well as the type of soil. The addition of olive oil cake @ 8% increased the organic matter content, water holding capacity, infiltration rate and soil water content of clay soil by 16.5, 10.3, 27.1 and 35 % as compared to control as well as its lower doses. While, in silt loam soil, the corresponding improvements were 4.68, 4.54, 34.5 and 13.28%, respectively as compared to control (Al-Widyan *et al.*, 2005). Supplementation of castor oil cake @ 6t/ha significantly decreased soil bulk density by 1.29% while increasing water holding capacity by 3.15% and infiltration rate by 0.50% over the control (Praveenkumar, 2000).

Soil chemical property

Castor oil cake significantly improves soil fertility by fundamentally altering both the chemical and physical properties of the rhizosphere, thereby promoting plant growth and development. According to the experiment conducted by Praveenkumar (2000), the application of castor oilcake @ 6 t/ha significantly influenced soil chemistry by decreasing soil pH by 0.76%, which enhanced the bioavailability of several essential nutrients. Concurrently, this application also increased the soil organic carbon by 0.48% and significantly enhanced the availability of macronutrients, specifically nitrogen (13.01%), phosphorus (12.47%) and potassium (9.46%) compared to control plots. The beneficial effects of castor oil cake extend beyond primary nutrients to essential trace elements. The study also recorded marked improvements in the availability of iron (17.6%), copper (26.17%), zinc (30.07%) and manganese (13.05%). These findings indicated that castor oilcake can not only be used as a fertilizer, but as a comprehensive soil conditioner that addresses the nutritional requirements of crops. The comparative efficacy of organic oilcakes against traditional inorganic fertilizers highlighted their superior ability to sustain soil vitality over time. Amani (2010) demonstrated that combined application of 100% Recommended Dose of Nitrogen (RDN) through castor cake resulted in higher levels of available nitrogen (241.26 kg/ha), sulphur (11.03 mg/kg) and organic carbon (0.77%) than standard NPK fertilizers. This organic approach also resulted in higher concentrations of manganese and zinc, suggesting that oilcakes contribute to a more balanced and diverse nutrient profile in the soil matrix.

Similarly, other oilseed by-products have shown comparable improvements in soil chemical properties ultimately contributing to its restoration. The addition of

palm kernel cake @10 t/ha nearly doubled soil organic carbon and significantly raised soil pH from an acidic 4.23 to a more neutral 5.88 (Conteh *et al.*, 2021). Guimaraes *et al.* (2018) further supported these trends by observing that castor cake supplementation increased the availability of calcium, magnesium and organic matter availability, attributing these gains to the high organic matter content inherent in the cake itself.

Furthermore, the integration of oilcakes like neem and castor into traditional farming systems provides significant physical and structural advantages to the soil. Yadav *et al.* (2019) reported that combined application of neem cake @5 q/ha with NPK fertilizers in cluster bean cultivation reduced the soil pH from 7.33 to 6.76, while simultaneously increasing electrical conductivity (EC) and organic carbon content. This treatment enhanced the availability of nitrogen, phosphorus and potassium to levels of 324.26, 32.93 and 203.22 kg/ha, respectively over control environments. Beyond chemical enrichment, oilcakes improve the physical tilth of the land. Choudhary *et al.* (2020) reported that neem cake application reduced soil bulk density to 1.07 mg/m³ by increasing pore space to 49.01%. This enhancement in soil structure allows for better aeration and moisture retention and when coupled with the elevated nutrient levels, creates an ideal environment for sustainable crop production.

Soil biological property

The application of castor oil cake, either alone or in combination with organic liquid manures improved the soil microbial population (Shilpashree *et al.*, 2022). However, the combined application of castor oilcake and 3% foliar spray of panchagavya enhanced the populations of bacteria, actinomycetes, N fixers and phosphate solubilizing microbes and fungi in descending order at all growth stages of field bean. However, the increasing microbial population was found to show a decreasing trend after 60 days of sowing. Soil biological activity is largely governed by the availability of organic carbon sources, which stimulate microbial growth and enzymatic processes. The faster decomposition as well as nutrient mineralization from castor oil was associated to higher cumulative CO₂ released by soil microorganisms due to their greater microbial activity (Ruiz-Valdiviezo *et al.*, 2013). Castor oil cake also exerts a beneficial influence on soil fertility and microbiological properties. Its application in combination with a microbial consortium enhanced soil organic carbon, availability of phosphorus, sulphur and mineral nitrogen levels. It also improved the activities of dehydrogenase, alkaline phosphatase and acid fluorescein diacetate (Basak *et al.*, 2020). Studies also suggest that the combined application of 100% RDN through pongamia cake with balanced P and K through fertilizer reported the highest urease (88.47 µgNH₄ +- N g⁻¹ hr⁻¹), acid phosphate (33.25 µgPNP g⁻¹ hr⁻¹) and

alkaline phosphate (35.48 µg PNP g⁻¹ hr⁻¹) activity in soil (Amani, 2010).

Effect of castor oil cake on plant growth and development

Among the major nutrients, nitrogen plays the most crucial role in plant growth and development as it is an integral component of chlorophyll, amino acids, proteins, nucleic acids, ribosomes, genes and other cell organelles. Beyond structural importance, nitrogen is a core constituent of enzymes and genes. The nitrogen supplied by castor oil cake supports chlorophyll synthesis, protein formation and overall plant growth and development. Several studies have demonstrated the positive influence of COC on plant growth and productivity. Research by Shilpashree *et al.* (2022) demonstrated that the combined application of castor oil cake and a 3% panchagavya spray produced significantly higher yields compared to untreated controls. Similar beneficial effects extend to other oilseed by-products. For instance, supplementation with pongamia cake has been shown to improve yields to 25.1 q/ha, while its combination with panchagavya pushes that figure further to 31.6 q/ha. Other soil studies revealed that pongamia cake applied @ 7.5 t/ha results in peak nitrogen (260 mg/kg) and potassium (189 mg/kg) levels within 30 to 60 days of application (Shilpashree *et al.*, 2022). This helps establish a nutrient-rich environment during the vegetative phase of the crop. The morphological and phyto-chemical properties of high-value crops are seen to have improved by the application of castor oil cake. The application of COC has been associated with improvements in both morphological and phytochemical characteristics of crops. In different studies involving chilli, it has been observed that application of composted simarouba oil cake significantly improved growth and yield. In chilli, the application of composted simarouba oil cake was found to produce twice the fruit yield as compared to 100 % NPK treatments. It has shown a positive correlation between fruit length and vitamin C content (Vasudhaudupa *et al.*, 2022). Furthermore, the application of castor cake @1 t/ha combined with nitrogen @ 150 kg/ha improved growth and yield attributes in chilli over the treatments having only inorganic nitrogen (Kacha, 2008). Similar success has been observed in medicinal plants; for example, Basak *et al.* (2020) reported that applying castor cake @2.5 t/ha alongside a microbial consortium (5 ml/kg seed) led to a 16.9% increase in fresh root yield and a 29.3% increase in dry root yield for Ashwagandha (*Withania somnifera*). These improvements were accompanied by notable gains in root length and girth, measuring 19.5% and 11.5% respectively over the control groups. Another experiment revealed that combining castor oil cake with biofertilizers and recommended dose of fertilizers have enhanced leaf sugar content and nutrient uptake (Chittibabu, 2008). Additionally, Praveenkumar (2000) observed that nitrogen-augmented castor cake application resulted in the highest nutrient recovery in carrot

roots. The nutrient uptake increased up to 69.78 kg ha⁻¹ for nitrogen and significant concentrations of micronutrients like zinc, iron, manganese and copper. This overall improvement in plant vigour often contributes to better disease resistance. The mustard oil cake combined with vermicompost maximized dry matter (24.15%) and total soluble solids in potato (Singh and Sharma, 2024). While, neem cake was effective in reducing root-knot nematode infestation in soybean (Mishra and Pandre, 2017). Thus, the consistent use of castor oil cakes serves as a sustainable strategy to boost crop quality and yield while fostering a beneficial microbial ecosystem in the soil (Shilpashree *et al.*, 2022).

Relevance of castor oil cake to sericulture host plants

COC contains substantial amounts of nitrogen, phosphorus, potassium, organic matter and several micronutrients which helped in improvement of soil fertility (Gadhesariya and Malam, 2017). COC has the potential to combine organic nutrient management with host plant cultivation. India produces a substantial quantity of castor oil cake annually as a by-product of its large castor oil industry (approx. 1 to 1.2 MT per year) (The Economic Times, 2026), creating an opportunity to recycle nutrients and organic matter within castor-based production systems. Moreover, optimized utilization of COC helps to reduce dependency on expensive inorganic fertilizers. Although bulky organic manures help in soil fertility management, they generally contain relatively low concentrations of essential plant nutrients. Consequently, they are required in large quantities, making their transport and application expensive and often limiting their availability (Verma *et al.*, 2019). On the contrary, COC is found to provide higher nutrient concentrations at lower application rates thereby increasing the overall quality of the soil at a lower amount (Gadhesariya and Malam, 2017). Moreover, COC also exhibits additional biological benefits, including reported nematicidal and acaricidal properties and the potential to enhance beneficial soil microbial activity and biodiversity (Gahukar, 2017). Nevertheless, practical adoption requires careful consideration of ricin and ricinine toxicity, appropriate decomposition periods and standard application rates before the leaves are harvested for silkworm feeding. Therefore, detailed field and feeding studies are necessary to evaluate and establish the agronomic, ecological and sericultural benefits of COC under castor-based production systems. In earlier studies, it has been already documented that better soil fertility promotes vigorous castor growth and greater leaf production, which is especially important in eri sericulture where a continuous supply of high-quality leaves is required throughout the rearing period (Bora *et al.*, 2025). Since, in ericulture, the performance of the eri silkworms depends entirely on the quality of the castor foliage, as such the leaf quality of castor plants is directly related to the larval growth, cocoon development and silk

production. Therefore, agronomic management practices that could improve the biomass as well as biochemical compositions of the castor plants have considerable potential for application in sericulture, particularly in ericulture. From a sericultural perspective, castor oil cake (COC) is of high relevance particularly due to its nutrient-rich organic amendment generated from the same crop that serves as one of the primary host plants of eri silkworms. This creates a circular nutrient-use system in which the waste product or the by-product which is generated during the extraction process of castor oil cake can be recycled back into castor production system thereby reducing dependency on external inputs of fertilizers and improving the long-term viability of sericulture-based castor cultivation. Henceforth, COC has great potential to improve the productivity of host plants in organic and low-input cultivation systems. However, the available evidence remains largely indirect. There is little direct experimental evidence linking COC application with changes in castor leaf biochemical composition and the subsequent performance of eri silkworms highlighting the need for integrated agronomic and sericultural investigations. Majority of the published studies on COC have been conducted in crops like basil, wheat, peanut, onion, chili and tomato. More importantly, almost no study has simultaneously assessed COC application, castor leaf biochemical composition and silkworm growth and cocoon production within the same experimental framework. Consequently, the hypothesis that COC enhances eri silkworm productivity through improved host plant nutrition is biologically plausible but has not yet been experimentally validated.

FUTURE PROSPECTS

Castor oil cake (COC) has a considerable potential as an organic fertilizer, but the challenges need to be addressed for the large-scale utilization of COC in sustainable agriculture and ericulture. One of the main challenges is the standardization of detoxification because ricin and ricinine can remain in fresh castor oil cake and may have phytotoxic effects if applied before adequate decomposition (Annongu and Joseph, 2010). There is still a need for reliable and cost-effective methods of detoxification and decomposition for safe agricultural application.

Another significant challenge is the standardization of processing and application practices. The COC nutrient composition varies depending on oil extraction methods, seed decortication and processing conditions which results in inconsistencies in fertilizer quality and agronomic performance (Gupta *et al.*, 2004). Quality standards, moisture regulation protocols and validated decomposition periods will need to be established prior to planting to ensure predictable nutrient release and consistent crop responses. Furthermore, additional studies are required to establish crop-specific application rates and evaluate the long-term effects on soil health and nutrient dynamics.

CONCLUSION

The transition towards sustainability in agriculture and allied sectors necessitates the transformation from synthetic inputs to organic nutrient sources that improve both soil health and crop productivity. The castor oil cake (COC) has emerged as a valuable organic amendment for castor (*Ricinus communis* L), the primary host plant of eri silkworm (*Samia ricini*). Thus, the COC can effectively help in mitigating the economic burden of chemical inputs while fostering a robust microbial ecosystem, ensuring long-term viability and productivity of the Eri culture ecosystem for future generations.

Disclaimers

The views and conclusions expressed in this article are solely those of the authors and do not necessarily represent the views of their affiliated institutions. The authors are responsible for the accuracy and completeness of the information provided, but do not accept any liability for any direct or indirect losses resulting from the use of this content.

Conflict of interest

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